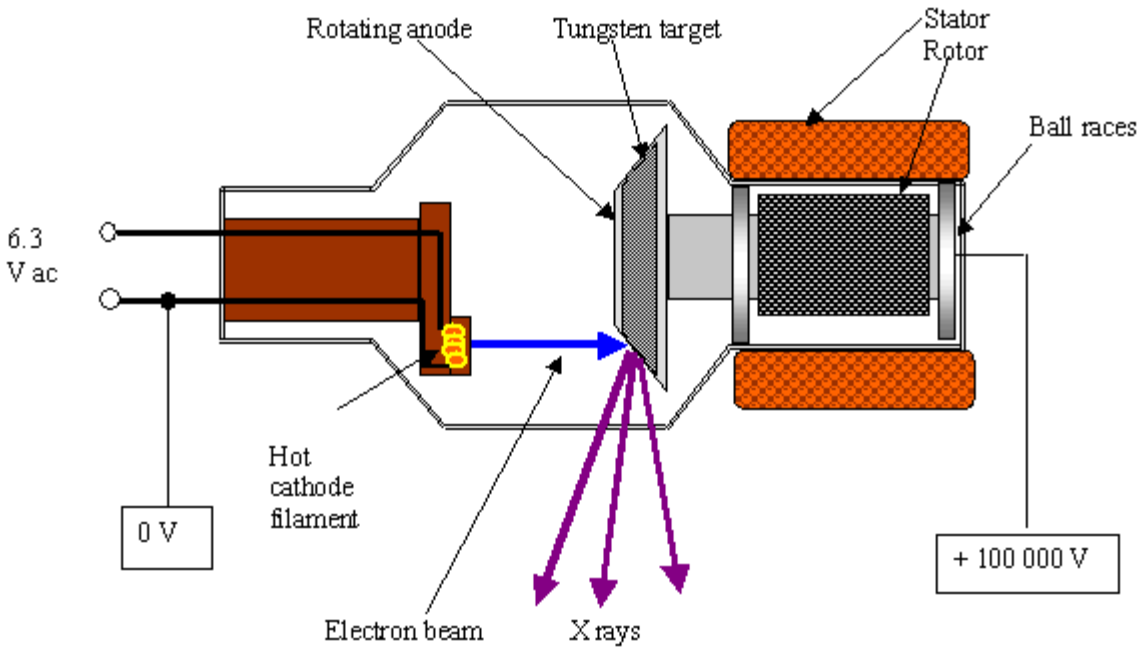
Radiology - introduction

Ya-Pei Chang, BVM, MVM, PhD, DipECVN, DAiCVIM (Neurology)

Production of X-ray

High-speed electrons strike metal ➔ x-rays produced

X-ray tube



mA: milliamperage

- Electric current passing through the filament → the number of electrons

mAs: milliampere seconds

- The amount of radiation produced by the x-ray tube

kVp: kilovoltage peak

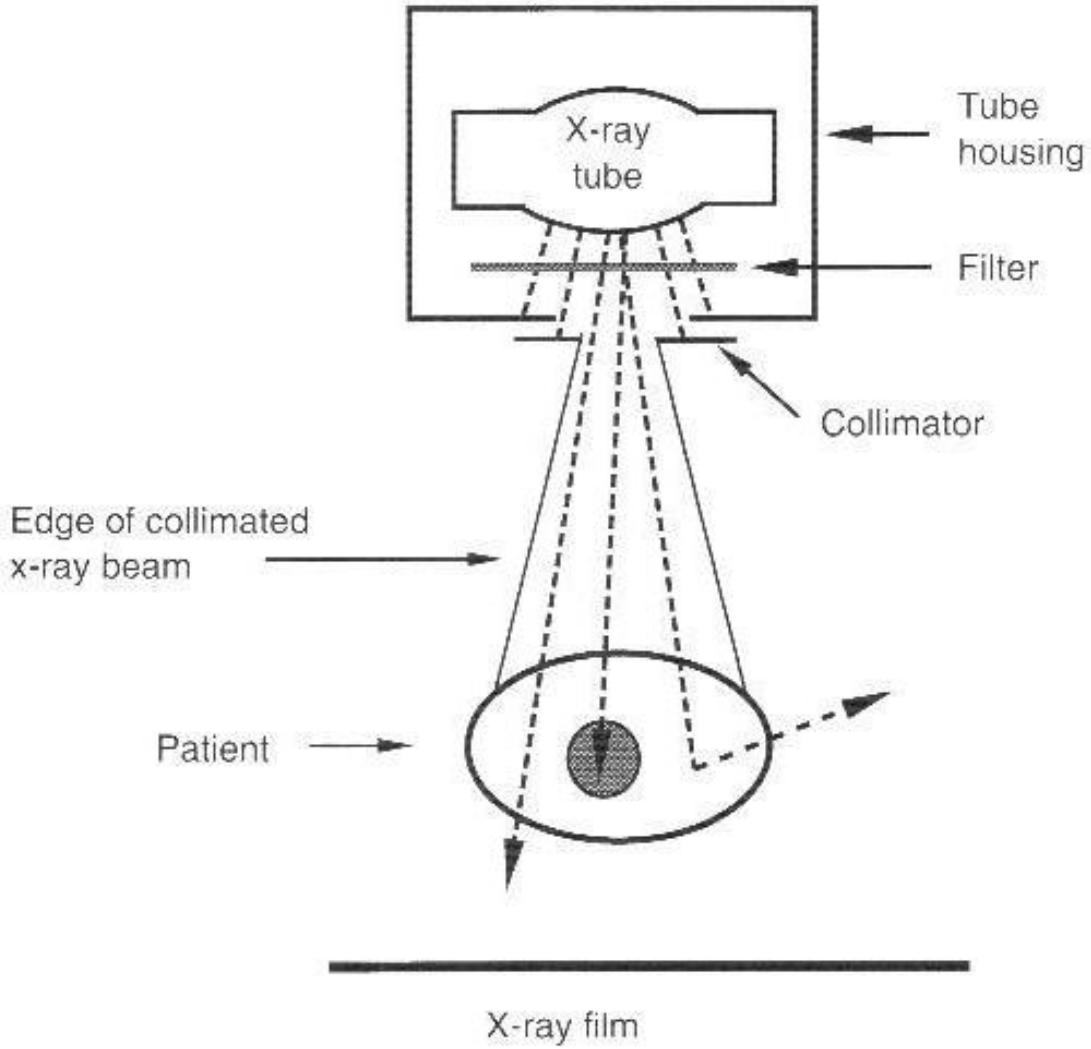
- Voltage difference between the anode and cathode
- ↑ kVp ⇢ electrons are accelerated to higher velocities ⇢ high-energy x-ray

Interaction of radiation with matter

- Many mechanisms

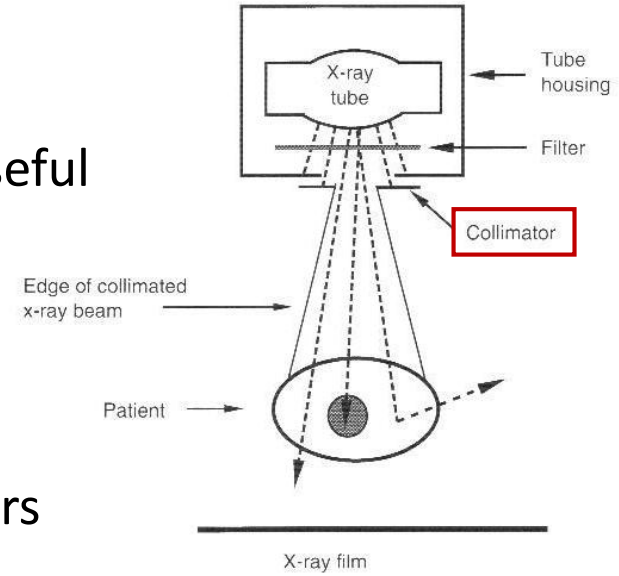
- Photoelectric effect (photoelectric absorption)
  - The main interaction process that results in an image
  - The X-ray striking the patient ⇢ totally absorbed by the patient ⇢ less x-ray exposure of the film
  - Proportional to Z^3 (Z = atomic number)
    - Ex. fat 6.5, bone 12.3, lead 82.0
  - Proportional to $1/E^3$ (E = photon energy)
    - ↑ energy of x-ray ⇢ ↓ photoelectric absorption ⇢ ↓ contrast between tissues of various types

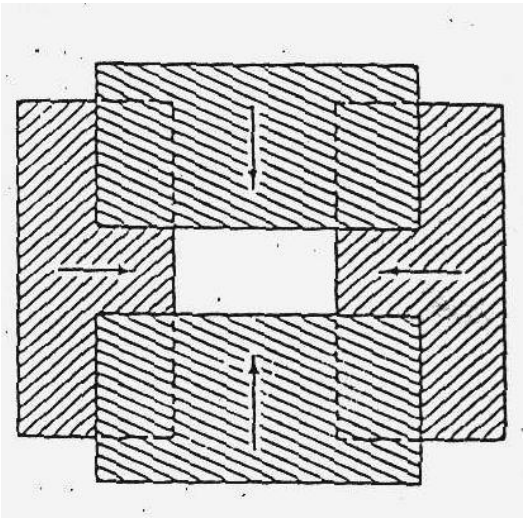
Basic concepts of making a radiograph


X-ray
tube
Tube
housing
Filter
Collimator
Edge of collimated
x-ray beam
Patient
X-ray film

Collimator / Collimation

- To limit the primary beam and prevent non-useful radiation from leaving the tube housing
- Non-useful radiation:
  - Increase patient dose
  - Degrade image quality because of fogging
    - Increase the radiation dose to bystanders







Film blackness / opacity

- Film darkening: directly related to the number of x-rays that reach that part of the film from the patient
  - ↑ degree of film darkness
    - ↑ X-ray machine output (mAs) ⇢ ↑number of x-rays striking the film
    - ↓ photoelectric absorption (kVp) ⇢ ↑x-ray exposure to the film

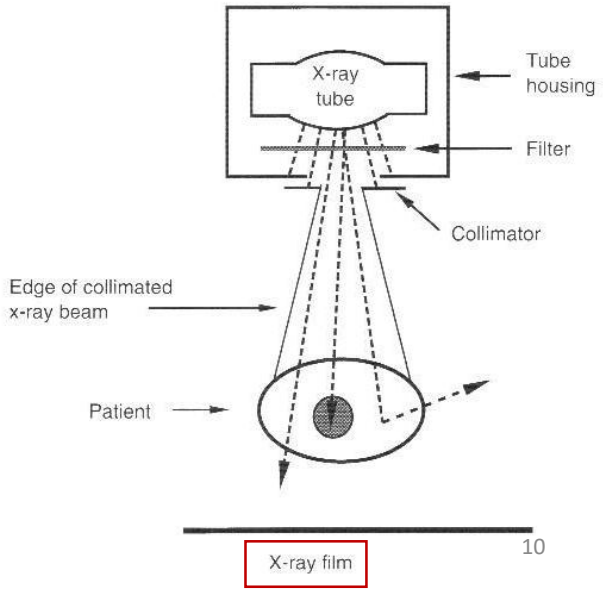
- X-ray film: light-sensitive emulsion containing silver halide

  X-rays or visible light

  ↓ sensitize silver halide

  ↓ *film processing*

  silver deposits, black in processed film



Factors affecting contrast

Radiographic contrast depends on:

- Film contrast: pressure, temperature, light, time
- Fog and scatter
- **Subject contrast**, affected by:
  - Physical density differences
  - Atomic number differences
  - Thickness differences
  - X-ray beam energy (kVp)

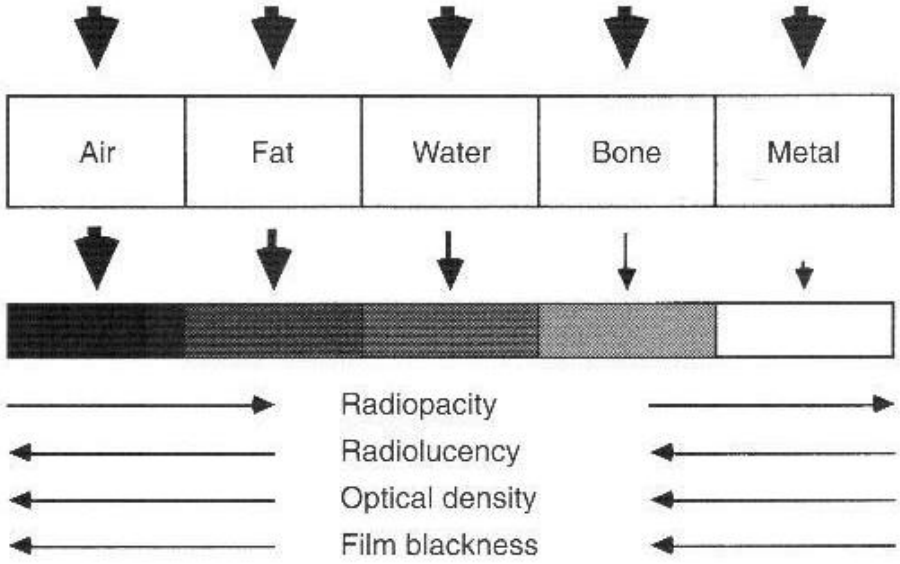
Physical density and atomic number of various substances

Substance	Physical density	Atomic number
Air	0.001	7.8
Fat	0.92	6.5
Water	1.00	7.5
Muscle	1.04	7.6
Bone	1.65	12.3
Lead	8.7	82.0

Fluid:
blood, urine, bile, CSF, transudates, exudates

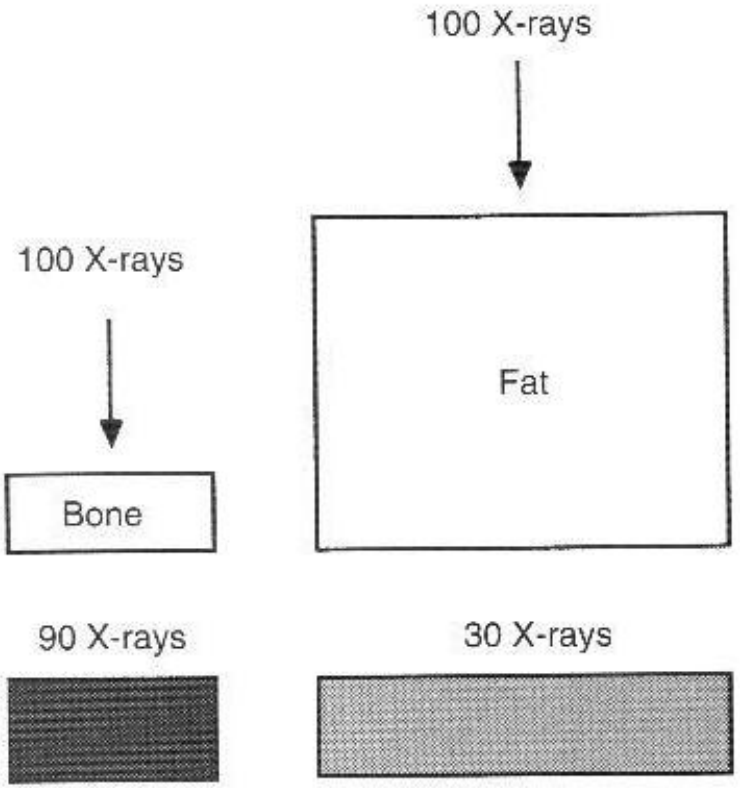
Tissue:
cartilage, muscle, fascia, tendons, ligaments, organs

Similar radiopacity
"Soft-tissue radiopacity"



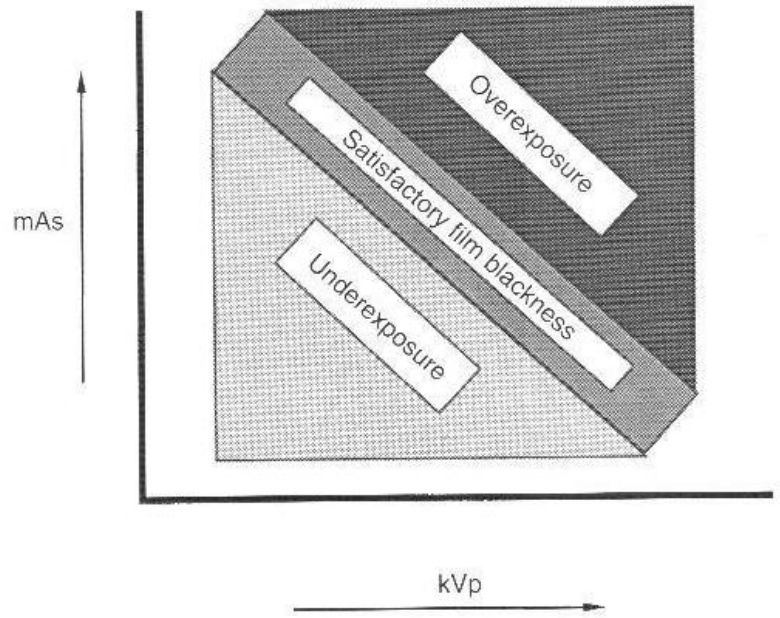
Thickness

↑ Thickness ⇢ ↑ radiopacity



X-ray beam energy

- ↑ kVp = ↑ x-ray energy
- kVp too low = too few x-rays penetrate the patient
  radiograph is underexposed
- kVp too high = too many x-rays penetrate the patient
  radiograph is overexposed



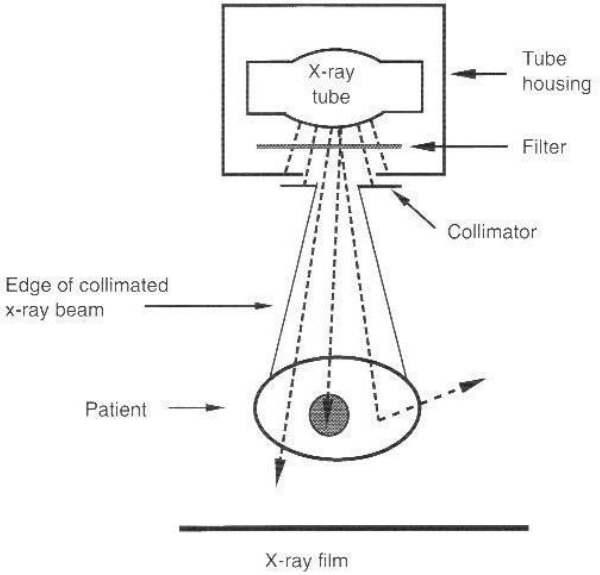
How to choose the kVp and mAs?

- Body part: ex. chest vs abdomen
- Characteristics: ex. motion

Consequences of radiographs being two dimensional…

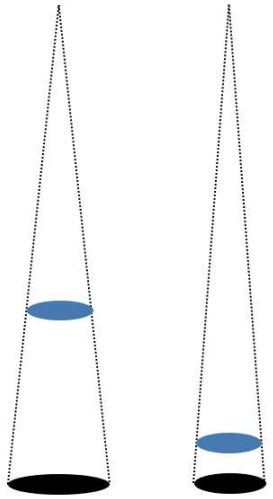
- Magnification and distortion
- Unfamiliar image
- Loss of depth perception
- Summation sign
- Silhouette effect / border effacement

Magnification and distortion



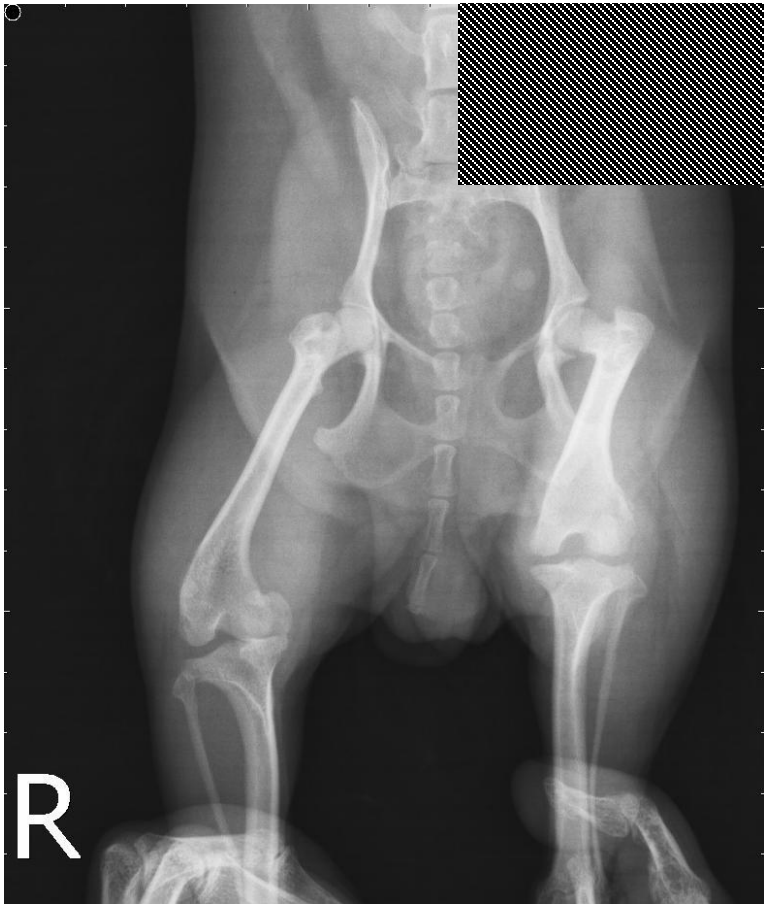
- ↑ Object-film distance ⇢ ↑ magnification
- The area of primary interest:
  - Placed closest to the cassette to avoid magnification
  - Exception:

    Lesion in the lung: the border of the lesion can be effaced by the atelectatic dependent lung



Distortion:

Unequal magnification of different parts of the same object

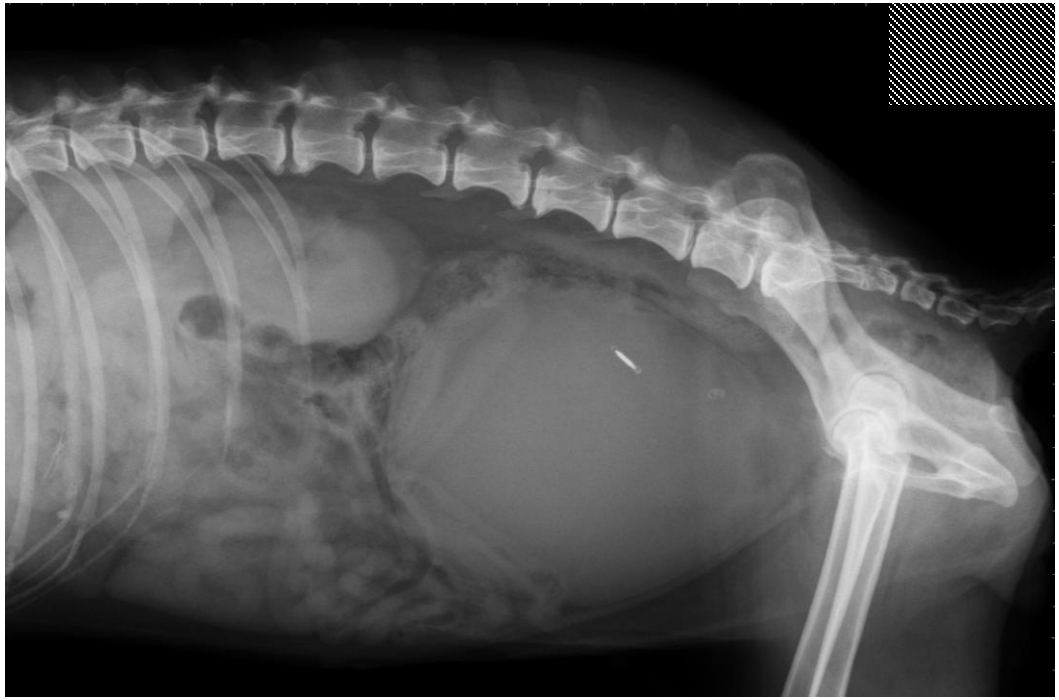


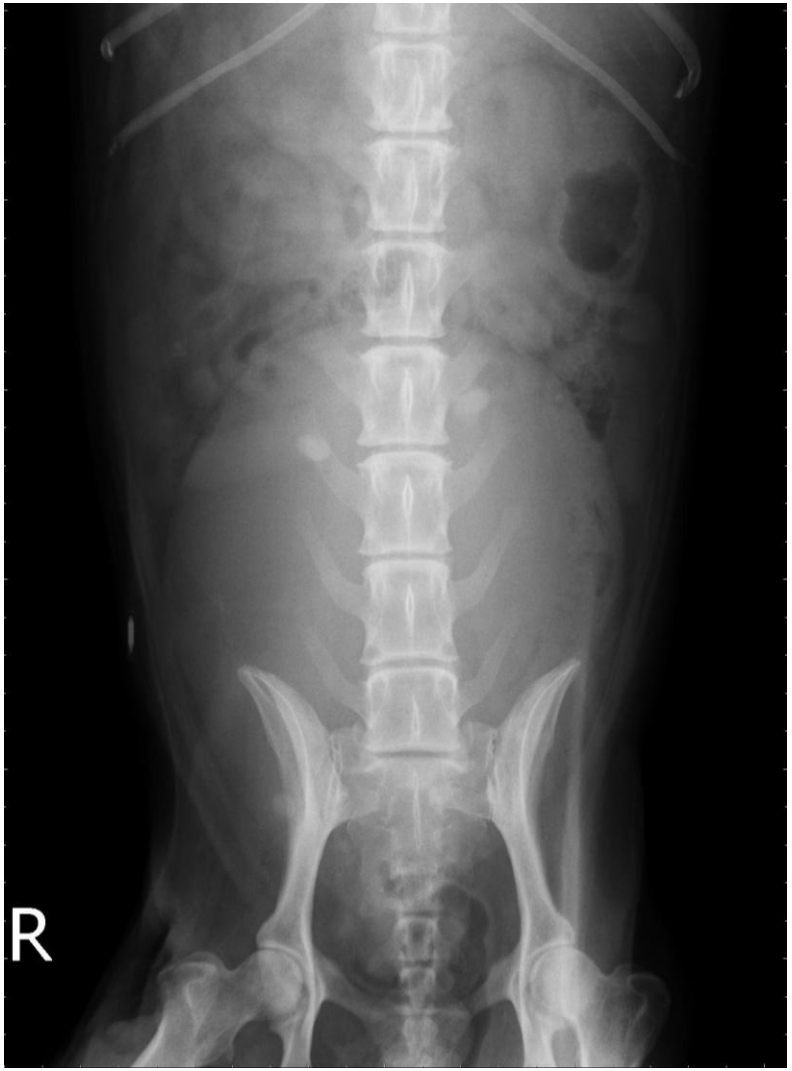
Unfamiliar image





Loss of depth perception





Summation sign

- The degree of x-ray absorption by all superimposed objects

Positive summation shadow



Negative summation shadow

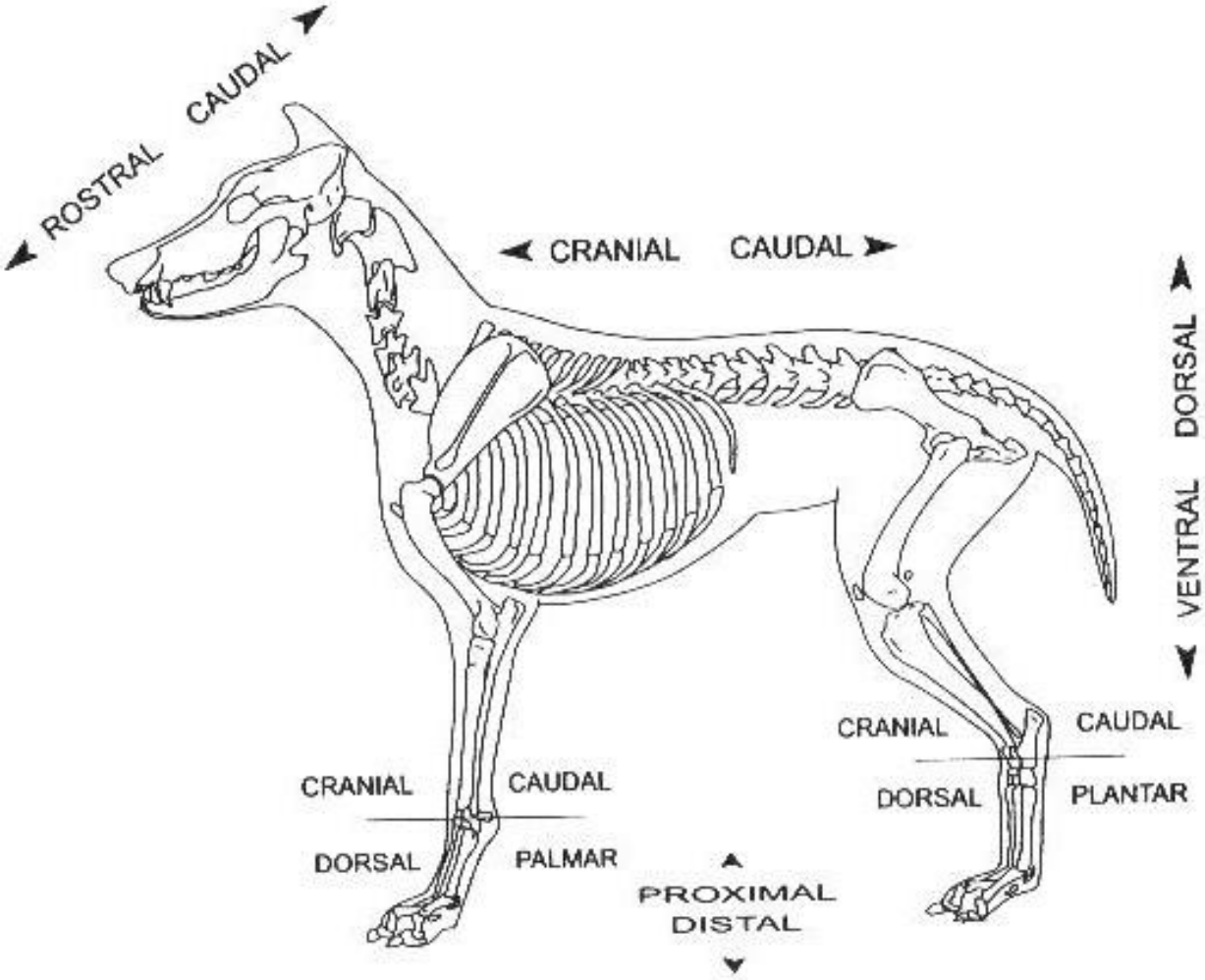


Silhouette effect / Border effacement

- When two structures of the same radiopacity are in contact, leading to the inability to distinguish their margin (border effacement)
- Conversely, if two structures of the same radiopacity are separated by a substance of a differing radiopacity, their borders can be distinguished radiographically



Radiographic projections and positioning


ROSTRAL
CAUDAL
CRANIAL
CAUDAL
VENTRAL
DORSAL
CRANIAL
CAUDAL
DORSAL
PALMAR
CRANIAL
CAUDAL
DORSAL
PLANTAR
PROXIMAL
DISTAL

Radiographic projection

- Named according to the direction of x-ray beam penetrates the body part of interest, from point-of-entrance to point-of-exit

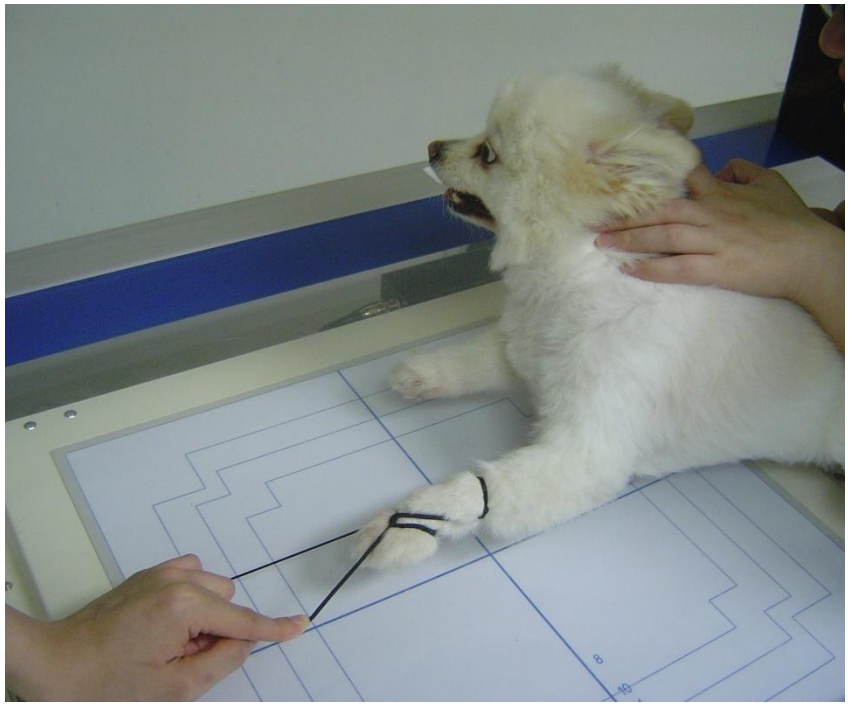
DV = dorsoventral projection
Animal in ventral recumbency



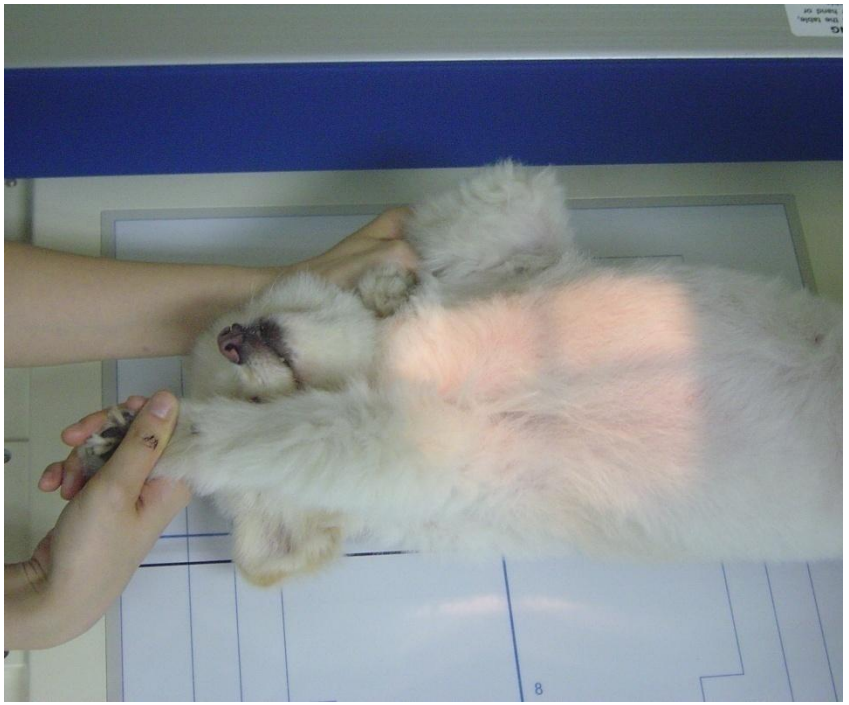
VD = Ventrodorsal projection
Animal in dorsal recumbency



Craniocaudal projection



Caudocranial projection



- Lateral radiographs of the abdomen and thorax
  - Named relative to the recumbency of the patient lying on the x-ray table, or in a standing patient, relative to the side of patient closest to the cassette
  - Right lateral projection Animal in right recumbency
  - Left lateral projection Animal in left recumbency

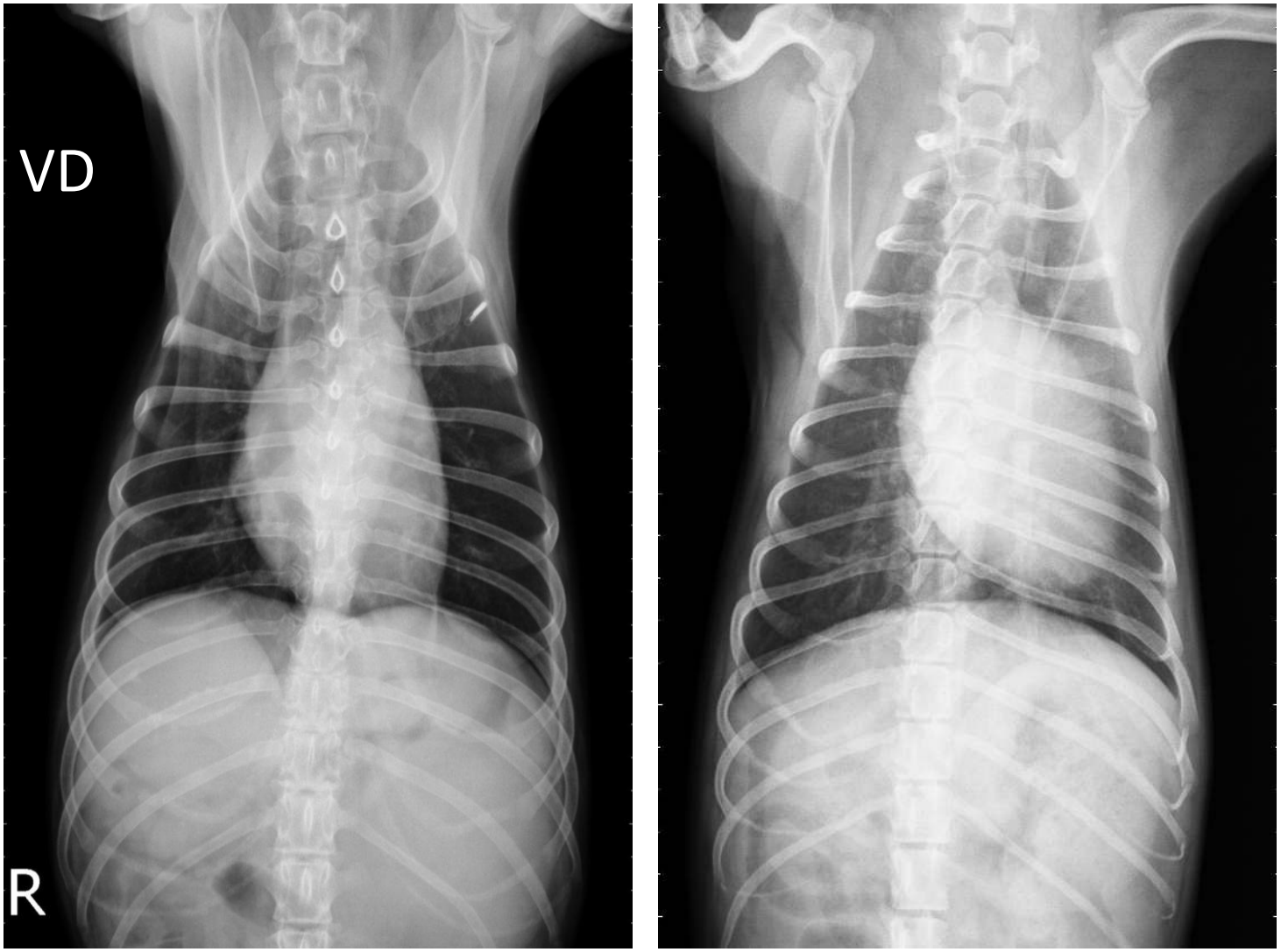


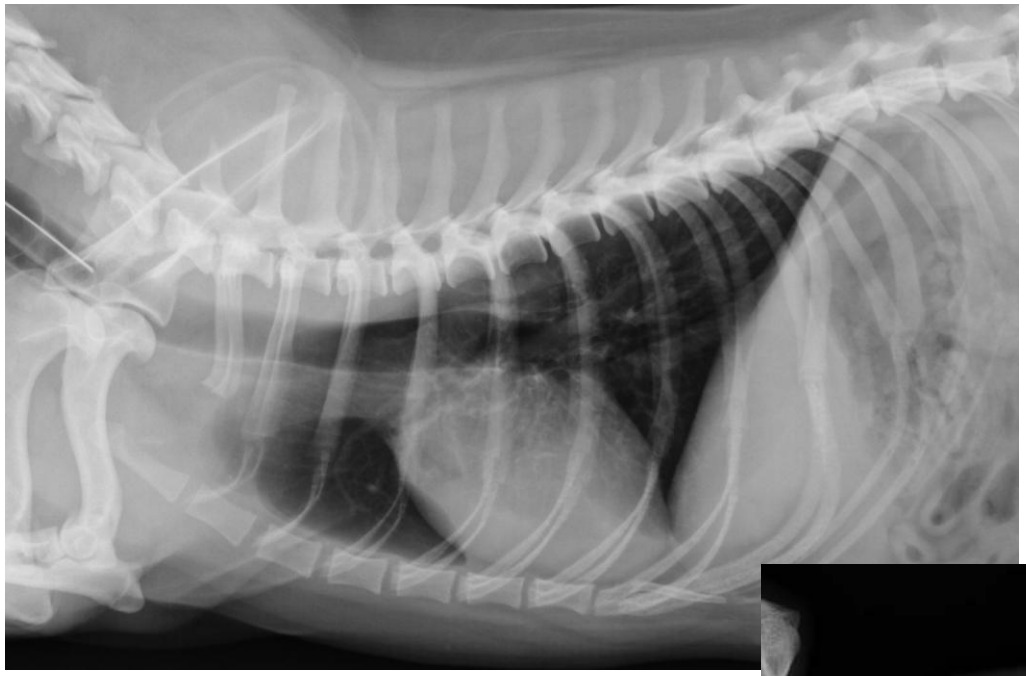
Right lateral projection

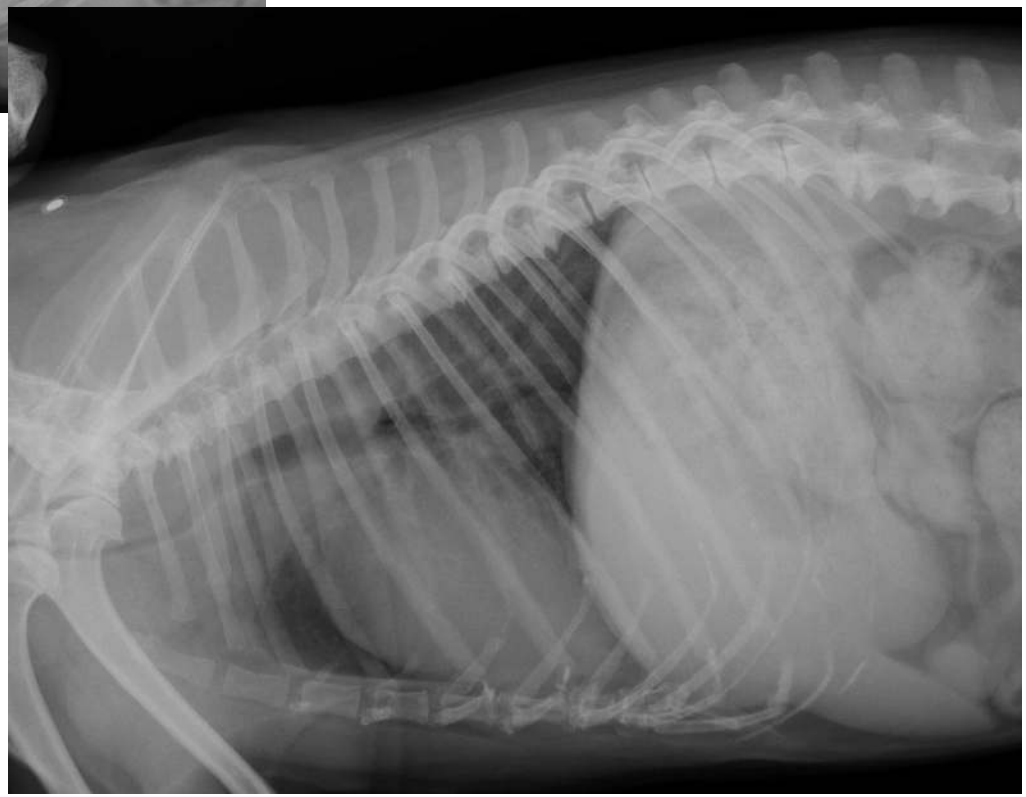


- Many other projections for different body parts
- Ex. TMJ, tympanic bulla, frontal sinus, hip, spine….

Importance of standard position







Procedures of taking a radiograph

- Choose the body part of interest
- Choose the appropriate projection (a minimum of two views 90 degrees to each other)
- Reduce objects overlapped with the body part of interest
  - External objects: collar, ET tube, monitoring equipment, fingers…
  - Patient: limbs, scapula, skin fold…

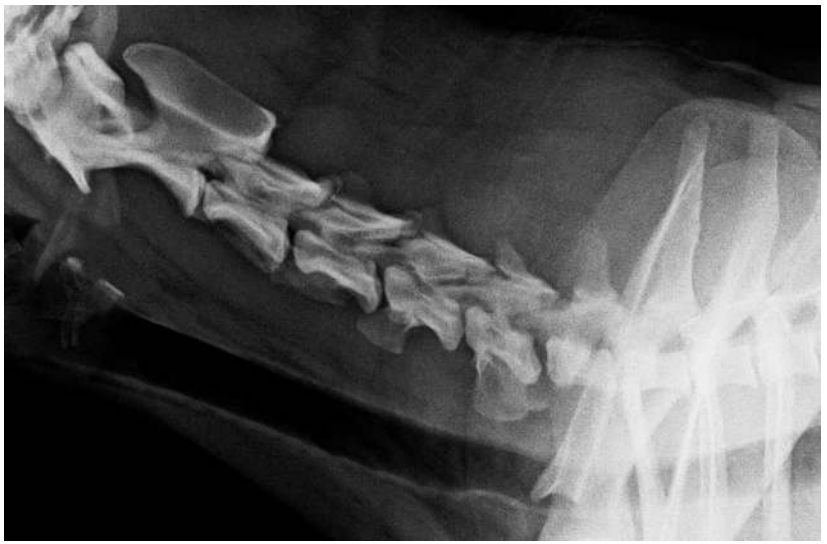

R


L

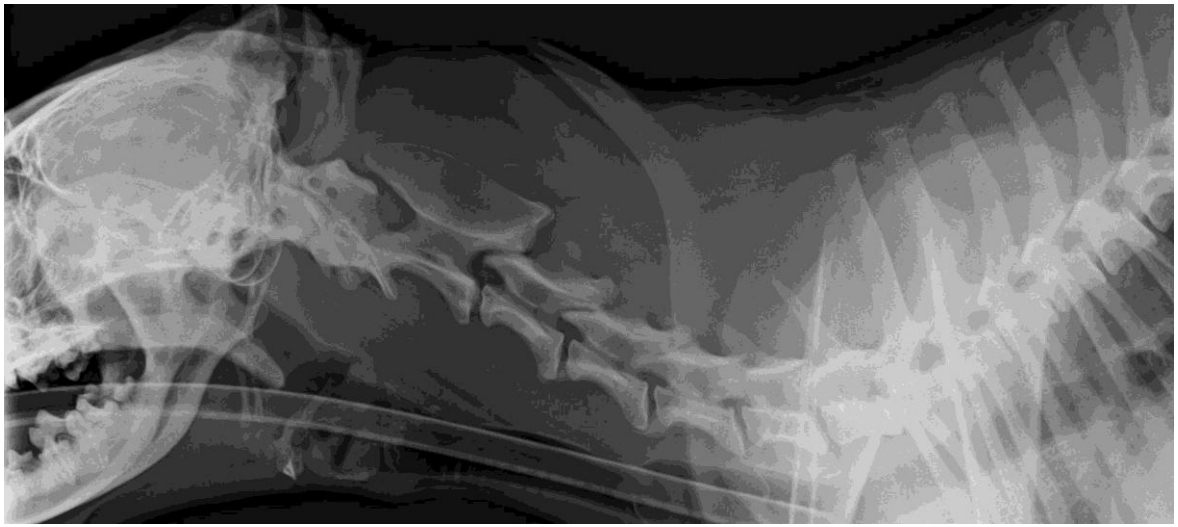


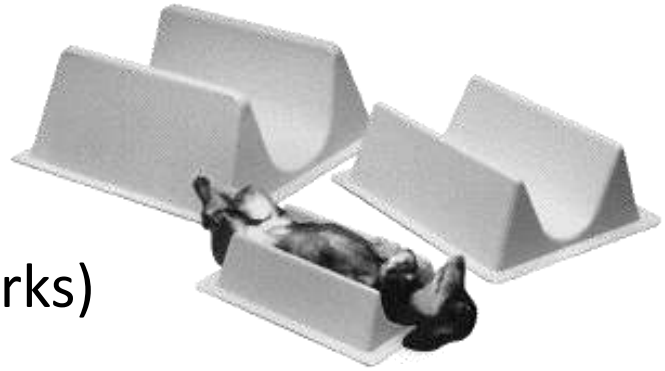
Scapula vs. caudal cervical spine



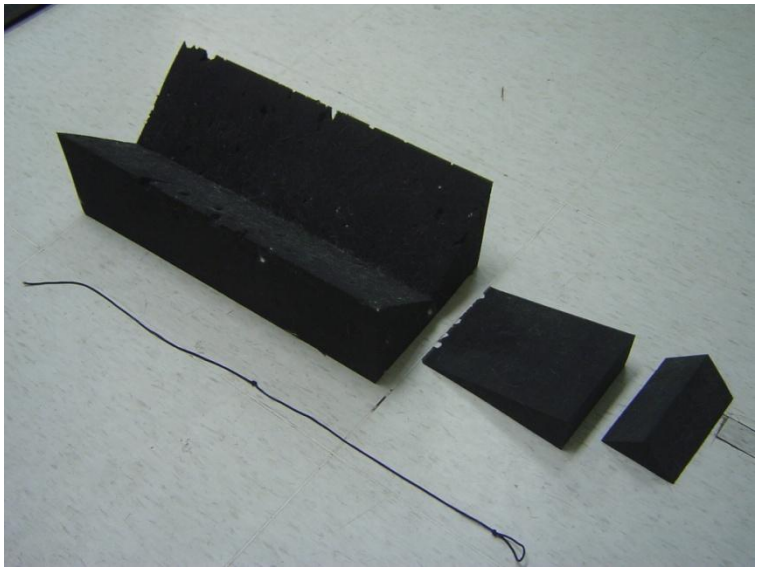


Skin fold





- Standard body position (use body landmarks)
- Effective patient restraint
  - Chemical: sedation, general anesthesia
  - Mechanical restraint devices: sandbag, foam cushing, tape, rope





- X-ray exposure area, collimation
- Other specific techniques and timing
- X-ray setting
  - Check the thickness of the target area
  - Decide the kVp and mA (mAs)